



Russ Garcia
Menlo Microsystems, Inc.
49 Discovery, Suite 100
Irvine, CA 92618
www.menlomicro.com

March 14, 2025

Faisal D'Souza, NCO
Office of Science and Technology Policy
Executive Office of the President
2415 Eisenhower Avenue
Alexandria, VA 22314

Submitted by email to [REDACTED]

Re: Request for Information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan ("Plan")

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

Menlo Microsystems ("Menlo Micro") strongly agrees with the Trump Administration that artificial intelligence (AI) will have revolutionary applications in economic innovation, job creation, national security, healthcare, and countless other industries, creating a new era of American excellence. However, existing problems in the supply chain and U.S. energy infrastructure threaten to restrict the development of AI in ways that could paralyze American innovation. It is apparent that we need to start investing in and building the infrastructure of tomorrow. For this reason, we respectfully submit the following proposal to the Office of Science and Technology Policy.

Where We Stand

The AI industry is not waiting for a lagging government or uncertain investors to catch up. With the race for AI dominance happening now, these advanced AI chip manufacturers will flock to



any component that meets its needs, including in adversarial countries willing to heavily invest in these critical solutions.

To counteract this, America must act decisively, focusing on foundational component technologies that enable AI manufacturing and infrastructure technologies at scale. We know the United States has been the most successful when it moves early to secure all aspects of innovative and disruptive designs. Just as we did in the Cold War, America should once again invest heavily in the innovations and components that will achieve its “uncontested technological superiority.”¹

There are several newer microelectronics technologies, including those using novel materials or technology platforms, that could result in the reduced size, weight, power and cost that will revolutionize not only the AI industry but also U.S. defense systems. In fact, DOD has already been directed by Congress in the 2025 National Defense Reauthorization report to foster, grow, and collaborate with domestic technology innovators on the development of these key components.² Rather than starting these investments from scratch, U.S. AI policy could instead build on existing DOD spending in dual-use AI component innovations.

To go a step further, DOD, the Department of Commerce, the National Science Foundation, the National Institute of Standards and Technology, the Nuclear Regulatory Commission, and domestic companies should all collaborate through additional public-private partnerships that prioritize funding and involvement in leap-ahead technologies.³ These investments will ensure U.S. technological superiority in AI and related fields.

Collaboration may include developing and sustaining a secure domestic supply chain and ensuring the protection of intellectual property, processes, manufacturing facilities, or other related technologies used in the manufacturing of such systems.

Strategic Investments for Long-term Technological Superiority

The quick development of DeepSeek’s latest AI model highlights the need for the United States to maintain its technological leadership and secure its intellectual property and domestic supply

¹ National Academies of Sciences, Engineering, and Medicine. 2024. Strategies to Enable Assured Access to Semiconductors for the Department of Defense. Washington, DC: The National Academies Press. <https://doi.org/10.17226/27624>. Page 52.

² Senate Armed Services Committee. July 8, 2024. National Defense Authorization Act for Fiscal Year 2025. <https://www.congress.gov/118/crpt/srpt188/CRPT-118srpt188.pdf>. Page 70.

³ National Academies of Sciences, Engineering, and Medicine. 2024. Strategies to Enable Assured Access to Semiconductors for the Department of Defense. Washington, DC: The National Academies Press. <https://doi.org/10.17226/27624>.



chain. We must prioritize the rapid deployment of advanced AI chips, relying on reliable switching solutions and improved manufacturing and testing processes to maintain leadership in this space.

While AI chips are critical, the broader ecosystem around semiconductor production—including key components—is equally important. Securing this entire supply chain is essential for sustaining technological advancement. The future of American AI and our technological autonomy rests on whether we are able to domestically produce essential components and address vulnerabilities in the supply chain.

Maximizing Energy Efficiency

It is no accident that the United States is developing cutting-edge AI infrastructure. The Department of Defense's commitment to nuclear energy, driven by the country's national security needs, in many ways has set America up for success in developing the energy resources required to meet this moment. However, there remain hurdles that the country must overcome to meet the growing energy demands from AI. While the private sector and governments look to expand and invest in energy resources, it is also important that energy be used as efficiently as possible to maximize the value of these new investments.

This will only be possible if the American government expands its investment in targeted advanced technologies. For example, by prioritizing energy efficient components in power plants and data centers, the government can secure an energy supply so vast that entire plants will not need to be built. Microgrids equipped with energy efficient technologies will be an excellent way to achieve this goal.

Protecting Critical Assets

Securing the AI supply chain both means promoting domestic manufacturing and protecting critical and expensive hardware at data centers, fabs, and power plants. Because heat poses considerable risks to these technologies, the government should prioritize investing in solutions that reduce, limit, or do not generate heat and mitigate dangers to these systems.

As the U.S. aims to maintain its technological leadership, we should invest in leap-ahead technologies that radically transform the manufacturing process, rather than simply evolving current technologies. Technologies like microelectromechanical systems (MEMS) switches and novel materials like glass substrates are critical in pushing the boundaries of AI chip



performance. These advanced MEMS devices actually outperform semiconductors in these types of applications due to their ultra-high conductivity, allowing better control of large amounts of electricity without producing added heat.

Glass substrates, too, have the potential to revolutionize semiconductor design by improving electrical performance, reducing thermal challenges, and enabling denser, more reliable circuits. While currently uncommon in the industry, prioritizing new innovations built on glass will position the United States far ahead of other nations and at the forefront of critical AI component technologies.

In fact, these low-heat solutions and novel materials will also set the United States up to succeed in technological revolutions that will follow AI—particularly quantum computing, which is thermally sensitive and requires energy solutions and electrical components that minimize heat. Investments in these innovations will lay the foundation for future breakthroughs and American technological excellence.

Conclusion

AI will drive the future of global economies, and the United States must prioritize the supply chain and energy infrastructure that will further American technological leadership.

Remaining competitive and securing our technological future requires the United States to significantly increase its investments not only in advanced chips, but in the other technological components that will make the existence of AI possible. Orienting these investments and public-private partnerships specifically towards leap-ahead innovations will ensure the United States maintains its global leadership in AI today and jumpstarts our leadership in the technologies of tomorrow.

About Menlo Micro

Headquartered in Irvine, California, Menlo Micro has created an entirely new category of electronic switches with its Ideal Switch® technology. The Ideal Switch® eliminates compromises and tradeoffs by combining the benefits of electromechanical and solid-state switches into the best of both worlds. Menlo Micro is bringing more than 99 percent reductions in size, weight, power, and cost to dozens of industries such as medical, aerospace and defense, telecommunications, consumer electronics, industrial IoT, and test and measurement.